

Experiment Report

Fraud Detection using MLflow Experiment Tracking

1. Objective

The objective of this project was to build a fraud detection system and implement a complete ML workflow using MLflow for experiment tracking, comparison, and model registry.

2. Dataset Description

Dataset: Fraud Detection Dataset

Source: <https://www.kaggle.com/datasets/kartik2112/fraud-detection>

Class 0: Legitimate Transaction

Class 1: Fraudulent Transaction

The dataset contains an imbalanced distribution of fraud and non-fraud transactions.

3. Data Preprocessing

Steps performed:

- Dataset loading
- Data cleaning
- Feature preprocessing
- Encoding categorical features
- Train-test split using stratification

4. Handling Class Imbalance

SMOTE was applied on training data to increase minority fraud samples and improve fraud detection performance.

5. Models Implemented

The following models were trained:

1. Logistic Regression
2. Random Forest
3. XGBoost
4. XGBoost + SMOTE

6. MLflow Experiment Tracking

MLflow was used to track:

Parameters:

- Model hyperparameters
- Sampling technique

Metrics:

- Accuracy
- Recall Class 0
- Recall Class 1
- F1 Score Macro

Artifacts:

- Trained model

7. Experiment Results

	run_id	tags.mlflow.runName	metrics.accuracy	metrics.recall_class_1	metrics.f1_score_macro
0	9f64314fc21142969af1ff95a7a958f3	XGBoost + SMOTE	0.987221	0.936945	0.726340
1	d4d9f92f4ef54e76a427411f20b7bb4e	XGBoost	0.998499	0.810835	0.930690
2	56b373965d484a3db5cf165971176e98	Random Forest	0.994221	0.001776	0.500324
3	c3a3cf2feafe48498420bf2898c79087	Logistic Regression	0.994211	0.000000	0.498549

8. Model Registry

The best performing model was registered in MLflow Model Registry.

Production Model:

anomaly-detection-prod

Aliases:

@challenger

@champion

9. Conclusion

The project demonstrates an end-to-end machine learning workflow including preprocessing, model training, MLflow tracking, model comparison, and production model registration.